

Does granting minority shareholders direct control over corporate decisions increase shareholder value? A natural experiment from China

Zhihong Chen,¹ Bin Ke,² and Zhifeng Yang³

Abstract

Using a unique 2004 Chinese securities regulation that requires equity offering proposals to seek the separate approval of minority shareholders, we show that giving minority shareholders direct control over corporate decisions increases the quality of corporate decisions and shareholder value in firms with higher mutual fund ownership. The regulation deters management from submitting value decreasing equity offering proposals, especially in firms with higher mutual fund ownership. Minority shareholders are more likely to veto value reducing equity offering proposals in the post-regulation period in firms with higher mutual fund ownership. The stock market reaction to the announcement of the regulation also increases with mutual fund ownership.

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¹ Assistant Professor of Accounting, City University of Hong Kong, 83 Tat Chee Avenue, Kowloon Tong, Kowloon, Hong Kong, China. Tel: +852 3442 6211. Fax: +852 2788 7944. Email: chenzhhh@cityu.edu.hk.

² Corresponding author and Professor of Accounting, Smeal College of Business, Pennsylvania State University, 316 Business Building, University Park, PA 16802, USA. Tel: +814-865-0572. Fax: +814-863-8393. Email: bxk127@psu.edu.

³ Assistant Professor of Accounting, City University of Hong Kong, 83 Tat Chee Avenue, Kowloon Tong, Kowloon, Hong Kong, China. Tel: +852 34424013. Fax: +852 2788 7944. Email: zhifeng@cityu.edu.hk.

1. Introduction

The expropriation of minority shareholders by management/controlling shareholders (hereafter referred to as management for brevity) is a worldwide corporate governance problem (Shleifer and Vishny 1997; La Porta et al. 1997, 1998; Djankov et al. 2008). Since minority shareholders in publicly traded firms usually delegate major corporate decisions to management, a common solution to management's expropriation is to design monitoring mechanisms (e.g., incentive compensation, board of directors, external auditors, etc.) to align the interests between management and minority shareholders. Due to the failure of many common monitoring mechanisms in controlling for management's expropriation of minority shareholders, there is a growing interest among activist shareholders in shifting the corporate decision making power from management to minority shareholders (see, e.g., Vascellaro and Tibken 2008). Government regulators around the world have also been busy introducing legislations to combat perceived managerial agency problems. In addition to enacting new laws to strengthen the effectiveness of existing monitoring mechanisms (e.g., the Sarbanes and Oxley Act in 2002), regulators are showing an increasing willingness to propose regulations that would grant minority shareholders direct control over corporate decisions (e.g., Scannell 2009; Ridley and Menon 2009).

However, whether minority shareholders should be granted direct control over corporate decisions is still hotly debated (see, e.g., Vascellaro and Tibken 2008; McCracken and Scannell 2009; Scannell 2009). Proponents (see, e.g., Bebchuk 2005) argue that granting minority shareholders direct control over corporate decisions is necessary to combat widespread

managerial agency problems and increase shareholder value.¹ Opponents (see, e.g., Bainbridge 2006) counter that minority shareholders' direct participation in corporate decisions reduces shareholder value because minority shareholders either lack the requisite knowledge and expertise to make effective decisions or have incentives to make value reducing decisions. For example, institutional investors, one major class of minority shareholders, are often accused of having a short term focus that induces management to make myopic decisions that hurt long term shareholder value (e.g., Porter 1992). There is also a possibility that increased minority shareholders' control over corporate decisions may force management to resort to alternative expropriation activities that are even more costly to minority shareholders.²

A priori it is difficult to quantify the costs and benefits of allowing minority shareholders direct control over corporate decisions. Hence, the debate on the consequences of granting minority shareholders direct control over corporate decisions will eventually have to be settled empirically. The objective of this study is to use a unique 2004 securities regulation issued by the China Securities Regulatory Commission (CSRC) to examine the effect of giving minority shareholders direct control over corporate decisions on the quality of targeted corporate decisions and shareholder value. Prior to the 2004 regulation management of publicly traded Chinese firms frequently expropriated minority shareholders using various mechanisms, including issuing new equity to minority shareholders. The new regulation intends to reduce the extent of

¹ In this paper shareholder value refers to the value of minority shareholders rather than the value of management, consistent with the existing governance literature (see, e.g., Shleifer and Vishny 1997).

² Harris and Raviv (2008a) offer an excellent analytic discussion on the costs and benefits of granting minority shareholders direct control over corporate decisions. See also Harris and Raviv (2005; 2008b) for related discussions.

management's expropriation by requiring several types of major corporate decisions (the most common of which is equity offering proposals) subject to the *separate* approval of minority shareholders. As detailed in Section 2, our unique setting allows us to overcome several common methodological challenges the extant literature faces in establishing the causal effect of changing minority shareholders' control over corporate decisions on shareholder value.

We conduct three types of complementary empirical analyses. Our first analysis uses the equity offering proposals over the period 1/1/2004-6/30/2005 as a specific setting to directly test whether the 2004 regulation has a deterrence effect by discouraging management from submitting value decreasing proposals. We use the stock market reaction to the announcement of an equity offering proposal (CAR) as a proxy for proposal quality. Despite the deterrence effect of the 2004 regulation, management may continue to submit value decreasing equity offering proposals in the post-regulation period. Hence, our second analysis examines whether minority shareholders are more likely to veto value decreasing proposals in the post-regulation period. Our final analysis uses the event study methodology to assess the overall stock market reaction to the announcement of the 2004 regulation.

The effect of the 2004 regulation on shareholder value should depend on the effectiveness with which minority shareholders exercise their newly granted control power. Hence, we also examine the influence of minority shareholder composition in each of the three empirical analyses. As small minority shareholders have a weaker incentive than large minority shareholders to exercise their voting rights, we focus on the top 10 minority shareholders. We

decompose the top 10 minority shareholders into institutional investors and individual investors because the former are often regarded as more sophisticated and better informed. We further decompose institutional investors into mutual funds and other miscellaneous institutions because mutual funds are more independent and thus should have a greater incentive to monitor firm management (Brickley et al. 1988; Chen et al. 2007).

Our primary results can be summarized as follows. We find that the 2004 regulation has a strong deterrence on value decreasing equity offering proposals, especially in firms with higher mutual fund ownership. There is no evidence that the 2004 regulation has any significant effect on management's likelihood of submitting value increasing equity offering proposals. Consistent with the 2004 regulation deterring value decreasing equity offering proposals, we find that the characteristics of the firms who submit equity offering proposals are more consistent with shareholder value maximization in the post-regulation period than in the pre-regulation period. In addition, the average CAR for the submitted proposals is significantly positive in the post-regulation period but significantly negative in the pre-regulation period. The difference in CAR across the two periods is significant.

In the post-regulation period we find no evidence that minority shareholders are more likely to veto value decreasing equity offering proposals on average. However, there is evidence that minority shareholders are more likely to veto value decreasing proposals in firms with higher mutual fund ownership but not in firms with higher other institutional investor ownership or individual investor ownership.

Consistent with the results on equity offering proposals, we also find that the stock market reaction to the announcement of the regulation increases with a firm's mutual fund ownership. We find no evidence that the stock market reaction to the regulation increases with other institutional investor ownership or individual investor ownership.

We conduct several robustness checks and find no evidence that our results are attributable to alternative explanations, such as other contemporaneous regulatory announcements or a general improvement in investor protection. Overall, our results suggest that giving minority shareholders direct control over corporate decisions can increase shareholder value when there are large and independent minority shareholders.

Our study makes several important contributions. First, we provide timely information to government regulators around the world who are debating about the costs and benefits of granting minority shareholders direct control over corporate decisions. The evidence from our study suggests that giving minority shareholders a direct say on corporate decisions does increase shareholder value, but only in firms with large and independent minority shareholders.

Second, we contribute to a growing international corporate governance literature following La Porta et al. (1997, 1998) that analyzes the effect of legal environment on shareholder value and financial market development. A general finding from this literature is that strong country-level investor protections are associated with higher shareholder value and faster financial market development (see, e.g., Djankov et al. 2008). Most studies in this literature examine a country's legal environment as a whole (typically using the indices from La Porta et al. 1998) and do not examine the specific mechanisms through which law affects financial markets.

In addition, the majority of the studies use cross-country regressions and therefore their conclusions are subject to the well-known concerns of endogeneity, measurement error, and correlated omitted variables (see La Porta et al. 2008). The contribution of our study is to use an exogenous regulatory shock to directly demonstrate how the adoption of one specific investor protection mechanism (i.e., the shift of corporate control from management to minority shareholders) affects the quality of corporate decisions and shareholder value.

Third, our results are also relevant to a growing U.S. literature on the proxy voting decisions of mutual funds (see, e.g., Davis and Kim 2007; Cremers and Romano 2007). The results from this literature are mixed, raising questions about the governance role of mutual funds. The evidence from our study suggests that a narrow focus on mutual funds' voting behavior alone would miss the deterrence effect of mutual fund ownership and thus would significantly understate the governance role of mutual funds.

Finally, our study also contributes to understanding the governance role of institutional investors in emerging markets with weak country-level investor protections. Despite the potential institutional frictions and agency conflicts that may hinder mutual funds' participation in corporate governance (e.g., Firth et al. 2010), our results suggest that mutual funds can play a positive role in strengthening the corporate governance of publicly traded firms in countries with weak investor protections.

The rest of the paper is organized as follows. Section 2 discusses the institutional background and related research. Section 3 discusses the effect of the regulation on management's proposal submission decision. Section 4 analyzes the relation between proposal

quality and minority shareholders' voting behavior. Section 5 presents the analysis of the stock market reaction to the announcement of the 2004 regulation. Section 6 concludes.

2. Institutional background and related research

2.1. Institutional background

Allen et al. (2005) show that mainland China is one of the worst countries in investor protections (defined as the sum of overall creditor rights, shareholder rights, rule of law, and government corruption). They find (see their figure 1) that China, along with Mexico and Indonesia, ranks the lowest in overall investor protections. In addition, mainland China lacks a free media, independent boards of directors (Fan et al. 2007), independent auditors (DeFond et al. 2000; Wang et al. 2008), and underdeveloped independent financial intermediaries such as financial analysts and institutional investors.

Prior to China's split share structure reform beginning in May 2005 that makes all shares tradable, domestically listed Chinese firms (often referred to as A share firms) had two types of common stocks: non-tradable shares, which are largely owned by the controlling shareholders (typically a local government, the central government, or an SOE), and tradable shares, which are listed on one of the two domestic stock exchanges and owned by Chinese citizens, domestic institutions and qualified foreign institutional investors. We refer to the tradable shareholders as minority shareholders in this paper. Except for the difference in tradability, the non-tradable

shares and tradable shares enjoy equal rights.³ Chen and Yuan (2006) find that the non-tradable shares owned by controlling shareholders of A share firms are very illiquid, thus limiting the ownership benefits of stock price appreciation to the controlling shareholders.

Because of the lack of investor protections and the illiquidity of non-tradable shares, controlling shareholders of A share firms have a strong incentive to use various means to tunnel the resources of A share firms to themselves (e.g., Jian and Wong 2008; Berkman et al. 2010; Fan et al. 2007; Jiang et al. 2010). Prior to the issuance of the 2004 regulation, management of A share firms often issued new equity to minority shareholders and then tunneled the proceeds of the equity offerings to controlling shareholders through various channels including related party transactions and related party loans (CSRC 2004). This phenomenon of malicious fund-raising was so widespread that it was dubbed “Quanqian” in Chinese.

To curb such egregious expropriation behavior, the CSRC issued a regulation in 2002 that required A share firms to seek the separate approval of tradable shareholders for any new share issuance that exceeds 20% of the firm’s total common shares outstanding. Unfortunately this regulation turned out to be ineffective because the 20% threshold is too high and most firms easily circumvented the regulation by simply issuing new equity below the 20% threshold. In 2004 the CSRC issued a tougher regulation that subjected several major corporate decisions (e.g., equity offering, major corporate restructuring, and overseas listing of subsidiaries) to the separate approval of tradable shareholders (often referred to as segmented voting rule). Equity offering is

³ Given that the majority of A share firms are controlled by the government, we also replicate all of our empirical results using state-controlled firms only and find similar inferences (untabulated).

the most common proposal subject to this regulation. The 2004 regulation was intended to be a temporary measure to combat widespread abusive equity offerings prior to the split share structure reform that would make non-tradable shares tradable. The 2004 regulation expired automatically upon the completion of the split share structure reform, which ended by the end of 2007 for most A share firms (see Li et al. 2008).

2.2. Related research

Minority shareholders in publicly traded firms typically delegate the control over corporate decisions to firm management but monitor the management's behavior through shareholder representatives, the board of directors. Therefore, the extant corporate governance literature primarily focuses on the design of effective control mechanisms (e.g., incentive compensation, board structure, etc.) to monitor firm management who makes corporate decisions. Much less is known from the extant literature on the economic effects of granting minority shareholders direct control over corporate decisions.

One stream of research relevant to us is the literature on shareholder activism. The common corporate issues targeted by activist shareholders include executive compensation, board structure, shareholder voting rights, and anti-takeover provisions in corporate charters (see, e.g., Johnson et al. 1997; Gordon and Pound 1993). Overall, this literature finds no conclusive evidence that shareholder activism has a significant impact on firm operations, earnings or stock returns (see Gillian and Starks 2007). As we discuss below, one reason for the mixed evidence is

the methodological challenges researchers face in establishing the causal effect of changing minority shareholders' control over corporate decisions on shareholder value (Gillian and Starks 2007).

Another stream of research relevant to us is the literature on the effect of U.S. state anti-takeover legislations in the 1980s on shareholder value (see, e.g., Bertrand and Mullainathan 2003; Garvey and Hanka 1999; Kim and Purnanandam 2009). As the state anti-takeover legislations increase minority shareholders' costs of intervening in firm management through the takeover threat, they could be viewed as a decrease in minority shareholders' control over corporate decisions. The evidence from this literature suggests that the state anti-takeover legislations increase managerial entrenchment and reduce shareholder value.

In response to the recent corporate scandals (e.g., Enron) and financial crisis, many governments have proposed regulatory rules that would grant minority shareholders an increased say on many important corporate issues such as executive compensation and director nomination. On April 20, 2007, the U.S. House of Representatives passed a Say-on-Pay Bill, which allows shareholders to have an annual advisory vote on executive compensation. Cai and Walkling (2010) find that the market reaction to the passage of the Say-on-Pay Bill was significantly positive for firms with high abnormal CEO compensation, with low pay-for-performance sensitivity, and responsive to shareholder pressure. Their evidence suggests that legislations that give shareholders a vote on executive compensation increase shareholder value for firms with inefficient compensation design and weak corporate governance.

Following La Porta et al. (1997, 1998), there is also a large international corporate governance literature that examines the cross-sectional association between country-level investor protections (defined as the sum of overall creditor rights and shareholder rights as specified in company and bankruptcy/reorganization laws, rule of law, and government corruption) and shareholder value and financial market development (see La Porta et al. 2008 for a review). The evidence from this literature suggests that strong country-level investor protections are associated with improved capital allocation (Wurgler 2000), higher shareholder value and faster financial market development (see, e.g., La Porta et al. 1997, 1998, 2002; La Porta et al. 2006; Djankov et al. 2008). This literature typically examines investor protections using the indices from La Porta et al. (1998) and does not examine the specific mechanisms through which law affects financial markets.⁴ Many studies in this literature also suffer from the problems of correlated omitted variables and endogeneity (see La Porta et al. 2008 for a discussion of these issues). We believe it is still an open question whether improving a country's investor protections would be sufficient to increase shareholder value. Furthermore, this literature generally does not distinguish investor protection provisions that facilitate minority shareholders' monitoring of management who makes corporate decisions from investor protection provisions that shift the control over corporate decisions from management to minority shareholders.

⁴ A notable exception is Atanasov et al. (2010) who study the effect of a 2002 Bulgarian law that strengthens minority shareholders' protection in several financial transactions (e.g., dilutive equity offerings, freezeouts and going-private transactions). However, they do not consider the effect of granting minority shareholders direct control over corporate decisions on shareholder value.

With respect to publicly traded Chinese firms, Berkman et al. (2010) examine the abnormal stock returns to the announcements of three Chinese securities regulations within a two-month period in 2000. The first regulation allows shareholders with more than 5% voting rights to propose motions for discussion at the shareholders' annual meeting and prohibits shareholders involved in a related party transaction from voting on the transaction. The second regulation prohibits listed firms from issuing loan guarantees to their shareholders, shareholders' controlled or affiliated companies, or any individual. The third regulation requires the board to perform a rigorous due diligence on any material asset acquisition or disposal. Berkman et al. find that firms with weaker governance experienced significantly larger abnormal returns around the announcements of the three regulations than did firms with stronger governance. While their results suggest that the three regulations help increase the degree of investor protections, it is difficult to determine whether the three regulations result in a significant increase in minority shareholders' direct control over corporate decisions.

There are several common methodological challenges that the extant literature faces in establishing a direct link between minority shareholders' control over corporate decisions and the quality of the targeted corporate decisions and therefore shareholder value. First, minority shareholders' control over corporate decisions often changes slowly. Therefore, a researcher may find it difficult to reliably measure a small change in minority shareholders' control or detect the effect of the small change on shareholder value. Second, most changes in minority shareholders' control over corporate decisions deal with general corporate governance issues (e.g., board structure or voting procedures) rather than specific corporate decisions. Hence, it is difficult to

directly attribute any observed change in managerial behavior (e.g., change in corporate investment) to a change in minority shareholders' control. Third, even if a change in minority shareholders' control deals with a specific corporate decision, a researcher generally cannot observe the outcome of the specific decision made by minority shareholders and thus has to infer the impact of the change in minority shareholders' control from aggregate performance outcomes such as stock prices or accounting earnings. As stock prices and accounting earnings reflect the effects of multiple economic forces, any association between changes in minority shareholders' control and changes in stock prices or earnings could be subject to alternative explanations (see Gillan and Starks 2007).

The experiment setting of our study can overcome all of these methodological challenges. In particular, the 2004 regulation represents a large shift in the control over corporate decisions from management to minority shareholders and deals with specific corporate decisions controlled by minority shareholders (i.e., equity offering proposals); in addition, we can observe the outcomes of those specific decisions. Therefore, it is relatively straightforward to draw the link between the increase in minority shareholders' control over corporate decisions and the change in the quality of the targeted corporate decisions in our setting.

3. The effect of the 2004 regulation on management's equity offering proposal submission decision

3.1. The sample

Although the 2004 regulation requires several types of managerial proposals (e.g., equity offering, major corporate restructuring, and overseas listing of subsidiaries) to be separately approved by tradable shareholders, we only use the equity offering proposals (including general offerings, rights offerings, and convertible bond offerings) for the following reasons. First, as noted in Section 2.1, equity offerings were one of the most common methods management employed to expropriate minority shareholders prior to 2004. Second, the frequency of equity offering proposals is considerably higher than that of any of the other managerial proposals. During our sample period the number of equity offering proposals is more than 200 but the number of the other types of managerial proposals such as the overseas listing of a subsidiary is less than a dozen and thus cannot be used to conduct a meaningful study. Third, for certain types of corporate proposals (e.g., major corporate restructuring), management can easily avoid the approval of tradable shareholders by manipulating the terms of the proposals. Hence, the sample of such proposals is severely biased. Finally, mixing different types of managerial proposals could create difficulty in identifying suitable control variables in our research design and the interpretation of our empirical results.

We limit our empirical analysis to the period 1/1/2004-6/30/2005. Our sample starts on 1/1/2004 because data on the detailed top 10 minority shareholder ownership are not available before 2004. Our sample ends on 6/30/2005 because very few firms submitted equity offering proposals for shareholder approval after June 2005, likely reflecting management's anticipation that the CSRC would not process equity offerings proposals due to the split share structure

reform.⁵ Note that all equity offering proposals approved by shareholders require the final approval of the CSRC.

We follow various CSRC regulations to identify all the A share firms that are eligible to propose equity offerings (general offerings, rights offerings, or convertible bond offerings) in a year (see Appendix A for the details of the identification method). There are 7,218 firm quarters during our sample period and 3,999 firm quarters (55%) are deemed eligible to propose equity offerings. The inferences in Table 2 are qualitatively similar if we include all of the 7,218 firm-quarter observations. We hand collected all the relevant information on the equity offering proposals submitted in our sample period, such as the announcement date, voting date, and the voting outcomes.

3.2. Methodology

For all the A share firms eligible to propose equity offering proposals in a quarter, we use the following multinomial logit model to test the effect of the 2004 regulation on management's decision to submit value increasing versus value decreasing equity offering proposals:

$$SUBMISSION_{it} = a + b * AFTER + c * CONTROL_{it} + \varepsilon_{it} \quad (1)$$

⁵ In fact most equity offering proposals announced in our post-regulation period were not processed by the CSRC due to the split share structure reform. As a result, we cannot compare how the proceeds from the equity offerings are used differently in the pre- and post-regulation periods.

where i and t are firm and quarter indicators, respectively. We estimate the model using quarterly data over the period 1/1/2004-6/30/2005 (6 quarters) because A share firms are allowed to submit equity offering proposals every quarter. In addition, the post-regulation period covers only two quarters and thus it is not appropriate to estimate the model using annual data.

$SUBMISSION_{it}$ is 0 if firm i does not submit a proposal in quarter t , 1 if firm i submits a value increasing (i.e., $CAR > 0$) proposal in quarter t , and 2 if a firm i submits a value decreasing (i.e., $CAR \leq 0$) proposal in quarter t . CAR is the market adjusted cumulative abnormal return over the $[-2, +10]$ trading days around the proposal announcement date. The difference between the proposal announcement date and the proposal voting date is at least 20 trading days for all but one proposal. For this one proposal, the holding period of CAR is 9 trading days only that end in the day before the voting date.⁶

We extend the holding period of CAR to 10 trading days after the proposal announcement in order to fully capture the market's assessment of the proposal quality. Equity offering is a complex business decision and thus minority shareholders may need more time to digest the information included in the proposal and search for private information to evaluate the merits of the proposal. This is especially important in China because management usually does not provide detailed information on equity offering proposals. In addition, the Chinese stock market is dominated by small retail investors and there are not enough sophisticated institutional investors such as financial analysts or institutional investors who can help quickly impound into

⁶ We also remove from CAR the effect of two material confounding events (i.e., earnings and dividend news) that occurred during the CAR measurement window and find similar inferences (untabulated).

stock prices the value implications of an equity offering proposal. Consistent with this argument, Ma (2004) finds a significant drift in the Chinese stock market's reactions to announcements of many major corporate decisions, including equity offering proposals. Hence, we believe that an abnormal return measured over a longer period should better capture proposal quality.⁷ Nevertheless, inferences on our key variables of interest in Tables 2, 3, and 6 are robust to using a shorter [-2, +2] event window. The only exception is that the mean marginal effect of *DCAR*MUTUAL_OWN* in Table 6's voting regression (see Section 4) becomes insignificant (untabulated).

AFTER is a dummy variable that is equal to one for the quarters in the post-regulation period (i.e., on or after December 7, 2004), and zero for the quarters in the pre-regulation period. *AFTER* is coded zero for the proposals submitted prior to December 7 in the 4th calendar quarter of 2004. The proposals submitted on or after December 7 in the 4th quarter of 2004 are treated as proposals submitted in the 1st quarter of 2005. As the number of months for calendar quarter 4 of 2004 is approximately 2 while the number of months for calendar quarter 1 of 2005 is approximately 4, we include the number of months in each quarter as a control.⁸ *CONTROL* is a list of common determinants of equity offerings discussed below.

⁷ Prior China related event studies also use relatively long periods to measure abnormal returns (see, e.g., Fan et al. 2008; Berkman et al. 2010).

⁸ Upon the release of the exposure draft of the regulation on September 27, 2004, some firms might have attempted to avoid the final regulation by accelerating future equity offering proposals to the period 9/27/2004-12/7/2004. As a robustness check, we also define *AFTER* using September 27, 2004 as a cutoff and find similar inferences.

If minority shareholders have the incentive to use their newly granted control power to veto value decreasing proposals, management should be deterred from submitting value decreasing proposals (i.e., the coefficient on *AFTER* should be negative for value decreasing proposals). In addition, to the extent that they are rational, minority shareholders should not veto value increasing equity offering proposals and therefore we should not expect the 2004 regulation to have a deterrence effect on value increasing equity offering proposals (i.e., the coefficient on *AFTER* should not be negative for value increasing proposals).

To make sure that the coefficient on *AFTER* is not due to systematic differences in the characteristics of the sample firms across the two time periods, we follow existing corporate finance research (see, e.g., Jung et al. 1996; Berger et al. 1997; Myers 2003; Leary and Roberts 2009) by including the following common equity financing determinants (see Table 1 for variable definitions). *Q* is a proxy for investment opportunities. We expect higher *Q* firms to be more likely to raise equity capital. *CASH* and *CFO* are proxies for the availability of internal funds. Firms with higher *CASH* and *CFO* are expected to be less likely to raise equity capital. *LEV* is a proxy for debt capacity and financial distress. We expect higher *LEV* firms to be more likely to raise equity capital. *VOLATILITY* is a proxy for the financial distress risk. We expect firms with higher *VOLATILITY* to be more likely to raise equity rather than debt. *AR12* is a proxy for the inverse of information asymmetry or stock price overvaluation. We expect firms with higher *AR12* to be more likely to issue equity capital.⁹ *ASSETS* is the natural logarithm of total

⁹ We also used the 12-month raw return or both *AR12* and the 12-month market return and found similar inferences (untabulated).

assets at the beginning of quarter t . *ASSETS* is a proxy for the inverse of information asymmetry and also controls for potential size effects.

The efficacy of the 2004 regulation should hinge on whether and how minority shareholders vote on submitted managerial proposals. Hence, we also examine whether the effect of *AFTER* varies with a firm's minority shareholder ownership structure. Economic theory suggests that the incentive to participate in shareholder voting should increase with a shareholder's stock ownership. Thus, we focus on the stock ownership of the top 10 minority shareholders, which is required to be disclosed quarterly since the end of 2003. In addition, due to their economy of scale, information advantage, and high level of sophistication, institutional investors should be more likely to participate in the voting and make more informed decisions than individual investors.

Institutional investors in China include mutual funds (open ended or close ended), securities firms, national social security trust funds, insurance companies, foreign institutions, etc. Brickley et al. (1988) and Chen et al. (2007) argue that relative to other institutional investors who may have existing or potential business relations with the listed firms (e.g., insurance companies) or who may have non-value maximizing social objectives (e.g., national social security trust funds), mutual funds are more independent and thus should be more likely to monitor firm management. In addition, mutual funds should face a greater pressure from retail investors to increase the return on their invested capital. Hence, mutual funds are expected to

have a stronger incentive to exercise their newly granted control power to deter or veto value decreasing managerial proposals.

Based on the above discussion, we consider the following minority shareholder ownership variables all measured at the end of the fiscal quarter immediately prior to the proposal announcement date. *MUTUAL_OWN* is the total stock ownership (as a percentage of the total outstanding tradable shares) of all the open ended and close ended mutual funds ranked among the top 10 minority shareholders. *OTHERINST_OWN* is the total stock ownership (as a percentage of the total outstanding tradable shares) of all the other institutional investors ranked among the top 10 minority shareholders.¹⁰ *INDIVIDUAL_OWN* is the total stock ownership (as a percentage of the total outstanding tradable shares) of all the individual investors ranked among the top 10 minority shareholders.¹¹

3.3. Results

¹⁰ We do not break out foreign shareholder ownership because there were very few foreign investors during our sample period, which predated the launch of China's Qualified Foreign Institutional Investor Program.

¹¹ We do not further decompose each top 10 minority shareholder type (e.g., mutual funds) by investment horizon. This is because value decreasing equity offering proposals, if approved, would result in an immediate decline in stock prices. Therefore, both long-horizon and short-horizon independent top 10 minority shareholders would have an incentive to veto such proposals. The level of aggregate stock ownership by each top 10 minority shareholder type is very stable over time (the AR(1) correlation is always greater than 70%), even though the investment horizons of individual shareholders within each top 10 minority shareholder type could vary.

Table 1 shows the descriptive statistics for the relevant regression variables of model (1). Approximately 2.60% of the firm quarters proposed value increasing equity offerings while 3% of the firm quarters proposed value decreasing equity offerings. The median size of the equity offerings (defined as the proposed dollar value of an offering scaled by the average market value of the tradable shares during the 20 calendar days before the equity offering announcement) is not significantly different over the pre- and post- regulation periods (untabulated).

Among the top 10 minority shareholders, the mean mutual fund ownership is 4.6% of the total outstanding tradable shares while the mean stock ownership of all the other institutional shareholders is 6.3% of the total outstanding tradable shares. These percentages are economically meaningful but are much lower than the mean total institutional ownership in listed U.S. firms. The mean individual shareholder ownership (INDIVIDUAL_OWNS) is 2% of the total outstanding tradable shares, much smaller than that of MUTUAL_OWNS or OTHERINST_OWNS. This finding suggests that most individual investors are not large shareholders even though they dominate the Chinese stock market in terms of numbers.¹²

The small aggregate ownership of the top 10 institutional investors raises an interesting question on the effectiveness of these institutional investors as monitors. We believe this is not a concern in our setting because the equity offering proposals must be *separately* approved by the minority shareholders who participate in the voting under the 2004 regulation. We find in untabulated analysis that in the post-regulation period non-top 10 minority shareholders rarely

¹² All of our regression results are similar if we require each top 10 minority shareholder's ownership to be at least 0.5% of the total outstanding tradable shares.

participated in the voting of equity offering proposals while the majority of the top 10 institutional investors did actively participate in the voting. Hence, the small aggregate ownership of the top 10 institutional investors could still have a substantial impact on the voting outcomes of the equity offering proposals under the 2004 regulation.

Panel A of Table 2 shows the regression results of the multinomial logit regression model (2) for the value increasing equity offering proposals in column (1) and value decreasing equity offering proposals in column (2). Note that the reference group in both columns is always the firms that do not have any equity offering proposals in a quarter. The coefficients on the control variables are generally consistent with our predictions though not always significant. The only exception is the coefficients on CFO and VOLATILITY in column (2).

For the firms that announced value increasing proposals in column (1), the insignificant coefficient on AFTER suggests that there is no evidence that increased minority shareholders' control over corporate decisions affects management's likelihood of submitting value increasing equity offering proposals. For the firms that announced value decreasing proposals in column (2), the coefficient on AFTER is significantly negative, suggesting that management is less likely to submit value decreasing equity offering proposals in the post-regulation period. Overall, these results suggest that the 2004 regulation significantly improves the quality of the equity offering proposals by deterring management from submitting value decreasing equity offering proposals.

Panel B of Table 2 shows the results of model (1) that allows the coefficient on AFTER to vary with the top 10 minority shareholder ownership characteristics. As this regression model

is nonlinear, the interpretation of the interaction terms is complicated. As illustrated by Ai and Norton (2003) and Norton et al. (2004) for the case of logit and probit models with one interaction term, the marginal effect of an interaction term in nonlinear models cannot be evaluated by simply looking at the sign, magnitude and significance of the coefficient on the interaction term. Following Ai and Norton (2003), we derive the correct marginal effect of an interaction term for the multinomial logit model in Appendix B.¹³ One key insight from this derivation is that the sign, magnitude and significance of the marginal effect of an interaction variable in the multinomial logit model could vary across observations and does not simply depend on the sign of the coefficient on the interaction variable. For example, the marginal effect of the interaction variable could be significantly different from zero even if the coefficient on the interaction variable is zero. Following Norton et al. (2004), we graph the distribution of the correct marginal effects and associated z-statistics for each interaction variable over the entire range of predicted probabilities for all observations in Figure 1. In addition, we report the mean marginal effect and mean z-statistic of each interaction variable in Panel C of Table 2.

There is little evidence that institutional ownership affects management's likelihood of submitting value increasing proposals in the post-regulation period relative to the pre-regulation period. As shown in Panel A of Figure 1, for value increasing equity offering proposals, the marginal effects for AFTER*MUTUAL_OWN and AFTER*OTHERINST_OWN are never significant at the 10% two-tailed significance level. However, there is weak evidence that

¹³ The STATA codes for computing the marginal effect of an interaction term in multinomial models are available from the authors upon request.

management of firms with higher individual minority shareholder ownership is less likely to submit value increasing proposals in the post-regulation period relative to the pre-regulation period. The mean marginal effect for AFTER*INDIVIDUAL_OWN is -0.4917 and close to be significant at the 10% level (mean z-statistic=-1.415). Though surprising, this latter finding is consistent with the allegation that increased individual minority shareholder participation in corporate decisions does not necessarily increase shareholder value.

For the firms that announced value decreasing proposals, the results in Panel C of Table 2 and Panel B of Figure 1 show that management of firms with higher mutual fund ownership (but not management of firms with higher other institutional investor ownership) is less likely to submit value decreasing equity offering proposals in the post-regulation period relative to the pre-regulation period. Specifically, as shown in Panel B of Figure 1, the marginal effect of AFTER*MUTUAL_OWN is significantly negative for many observations. The mean marginal effect of AFTER*MUTUAL_OWN is negative and significant at the 10% two-tailed level (mean z-statistic=-1.833).¹⁴ The marginal effect of AFTER*OTHERINST_OWN is insignificant at the 10% level for all observations.

With regard to top 10 individual minority shareholder ownership, the marginal effect of AFTER*INDIVIDUAL_OWN for value decreasing proposals is significantly negative for many observations. The mean marginal effect of AFTER*INDIVIDUAL_OWN is negative and

¹⁴ It is unlikely that the significant interaction effect for AFTER*MUTUAL_OWN can be explained by mutual funds' stock picking ability. If this were the case, the coefficient on MUTUAL_OWN should be significantly negative rather than insignificant for the pre-regulation period in column (2) of Table 2, Panel B.

significant at the 10% two-tailed level (mean z-statistic=-1.905). This result suggests that management of firms with higher individual minority shareholder ownership is less likely to submit value decreasing proposals in the post-regulation period relative to the pre-regulation period. However, since there is weak evidence that management of firms with higher individual shareholder ownership is also less likely to submit value *increasing* equity offering proposals, the net effect of the regulation on shareholder value is mixed for firms with higher individual minority shareholder ownership.

3.4. Quality of proposal submissions before and after the regulation

Given that the 2004 regulation helps deter value decreasing equity offering proposals as shown in Table 2, we should expect the average quality of the submitted equity offering proposals to be higher in the post-regulation period than in the pre-regulation period. We test this prediction in two complementary approaches. First, we examine whether the characteristics of the firms that submitted equity offering proposals are more consistent with shareholder value maximizing determinants of equity offerings in the post-regulation period than in the pre-regulation period. Existing finance theories (see, e.g., Jung et al. 1996) suggest that growth firms who face lower information asymmetry or have insufficient internally generated cash and low debt capacity should be more likely to propose equity offerings in order to increase shareholder value. Following the analysis in Section 3.2, we use Q as a proxy for growth, CASH and CFO as proxies for the availability of internal funds, LEV and VOLATILITY as proxies for debt

capacity, and AR12 and ASSETS as proxies for the information asymmetry between management and outside investors. We expect Q, LEV, VOLATILITY, AR12, and ASSETS to be higher in the post period while CASH and CFO to be lower in the post period. Second, we compare the stock market reactions to announcements of equity offering proposals (CAR) across the two time periods and expect the average CAR to be more positive in the post-regulation period.¹⁵

Panel A of Table 3 reports the descriptive statistics of the firm characteristics for the equity offering proposals submitted over the pre- and post- regulation periods. Consistent with our predictions, LEV and VOLATILITY are higher while CASH is lower in the post period. AR12 and ASSETS are indifferent between the two periods. Contrary to our prediction, Q is higher in the pre-regulation period. As the overall Chinese stock market is prone to the cycle of booms and busts, we also compute an adjusted Q by subtracting the mean Q for the whole market from a firm's raw Q. The adjusted Q is not significantly different for the two periods (untabulated). Overall, there is weak evidence that the firm characteristics of the equity offering firms are more consistent with value maximizing firms in the post-regulation period than in the pre-regulation period.

¹⁵ One limitation of CAR as a proposal quality proxy is that CAR may not be not very negative due to the stock market's anticipation of minority shareholders' veto of value decreasing proposals, even though management may continue to submit a large number of value decreasing equity offering proposals in the post-regulation period. However, this anticipation effect does not appear to be severe in our sample because as shown in Section 4, only a small number of equity offering proposals were vetoed by minority shareholders in the post period.

Panel B of Table 3 shows the results of CAR for the pre- and post- regulation periods. The results suggest that the proposal quality is higher in the post-regulation period. The mean and median CAR are both significantly negative in the pre-regulation period using the t-test and rank-sum test. In contrast, the mean and median CAR are both positive in the post-regulation period and significantly different from zero using the t-test. In addition, the mean and median CAR are significantly different over the two time periods using either a t-test or rank-sum test.

Panel C of Table 3 provides further evidence on the impact of the top 10 minority shareholder composition on the quality of submitted proposals across the two time periods.¹⁶ Since management had absolute control over equity offering decisions in the pre-regulation period, it may not be surprising to observe that none of the coefficients on `MUTUAL_OW``N`, `OTHERINST_OW``N`, and `INDIVIDUAL_OW``N` is significant. The coefficient on `AFTER*MUTUAL_OW``N` is significantly positive, suggesting that mutual fund shareholders play a more effective governance role in the post-regulation period by improving the quality of submitted proposals. There is no evidence of an improvement in proposal quality over the two time periods in firms with higher other institutional or individual shareholder ownership. Overall, Table 3's results are consistent with Table 2's results.

¹⁶ For both Panels B and C of Table 3, we do not control for the firm characteristics in Panel A because any improvement in CAR in the post-regulation period is due to the change in the types of firms that propose equity offerings. That is, CAR and the firm characteristics are two alternative tests of the same idea, even though it is likely the firm characteristics would not be as comprehensive as CAR in capturing the improvement in proposal quality. However, the results in Panels B and C are robust to controlling for the firm characteristics in Panel A (untabulated).

3.5. Robustness checks

We perform several sensitivity checks to rule out alternative explanations for the results in Tables 2 and 3. First, we check whether there are alternative government regulations issued during our 6-quarter sample period (1/1/2004-6/30/2005) that would *directly* constrain management's ability to submit value decreasing equity offering proposals. We find none after searching a comprehensive Chinese law database (www.chinalawinfo.com).

Second, we examine whether the results in Tables 2 and 3 could be explained by China's gradual improvement in investor protection during our time period rather than the segmented voting rule. This is a legitimate concern because the Chinese government has been steadily introducing various regulations and reforms to improve minority shareholders' protection, even though the enforcement of such regulations is always questionable. Even the 2004 regulation contains four additional minor investor protection provisions in addition to the segmented voting rule.¹⁷ We perform several types of analyses to rule out this alternative explanation. We first show in Figure 2 that there is no evidence of a time trend in the mean/median CAR over our sample period except for the jump in CAR coincident with the 2004 regulation. We also replicate the regressions in Panels A and C of Table 2 and Panels B and C of Table 3 using a pseudo

¹⁷ The four investor protection provisions are as follows: (a) strengthening the role of independent directors by requiring material related party transactions and the hiring and dismissal of the company auditor subject to the approval of at least one half of the independent directors; (b) improving investor relations by encouraging management to improve the quality of corporate disclosures and investor communications; (c) encouraging listed firms to adopt a regular dividend policy and prohibiting listed firms that have not distributed cash dividends in the past three years from issuing new equity; (d) holding controlling shareholders and company executives to the standard of fiduciary duty for minority shareholders and increasing the administrative penalties for violation of such fiduciary duty. Note that none of these four provisions *directly* constrain management's ability to propose equity offerings.

AFTER over the following three alternative 6-quarter time periods: (a) 2003Q1-2004Q2; (b) 2003Q2-2004Q3; and (c) 2003Q3-2004Q4. Similar to the definition of *AFTER*, the pseudo *AFTER* is one for the last two quarters and zero for the first four quarters for each of the three 6-quarter time periods (see Table 4).¹⁸ For brevity, Table 4 only shows the replicated interaction effects models in Panel C of Tables 2 and 3. For all the replications using the three alternative time periods, we find no results similar to those in Tables 2 and 3. Overall, these sensitivity results suggest that the results in Tables 2 and 3 are not due to a gradual improvement in investor protection.

Our second analysis examines whether there is a decline in the extent of inter-corporate loans from A share firms to their controlling shareholders in the post-regulation period. Jiang et al. (2010) show that inter-corporate loans are a common tunneling mechanism that controlling shareholders use to expropriate minority shareholders of publicly traded A share firms. While a general improvement in investor protection is expected to reduce controlling shareholders' incentive to expropriate minority shareholders using both inter-corporate loans and value decreasing equity offering proposals, the 2004 segmented voting rule does not directly limit management's ability to expropriate minority shareholders using inter-corporate loans. Therefore, to the extent that the results in Tables 2 and 3 are due to a general improvement in investor protection rather than the segmented voting rule, we should also observe a decline in the extent

¹⁸ As minority shareholder ownership data are unavailable before 2003Q4, we assume that the values of *MUTUAL_OW*N, *OTHERINST_OW*N, and *INDIVIDUAL_OW*N for the quarters before 2003Q4 are equal to those at the end of 2003Q4. This is a reasonable assumption because the top-10 minority shareholder ownership is very stable in our sample period (see footnote 11).

of inter-corporate loans post the 2004 regulation in our sample period. We use the following OLS regression model to test this prediction:

$$OREC_{it} = \beta_i + \beta_1 * AFTER + \beta_2 * LNTA_{it} + \varepsilon_{it} \quad (2)$$

where i and t are firm and quarter indicators, respectively. OREC (defined as gross other receivables deflated by year-end total assets) is a proxy for the inter-corporate loans per Jiang et al. (2010).¹⁹ AFTER is defined as in Section 3.2. Because OREC is scaled by total assets, we include LNTA (defined as the natural logarithm of year-end total assets) to control for size effects. The model includes firm fixed effects, but inference is similar without the firm fixed effects. To determine whether minority shareholder composition affects the coefficient on AFTER, we also allow the coefficient on AFTER to vary with MUTUAL_OWNS, OTHERINST_OWNS, and INDIVIDUAL_OWNS.

Table 5 shows the regression results of OREC for all publicly traded A share firms over our 6-quarter sample period 1/1/2004-6/30/2005. To avoid alternative explanation resulting from a change in the mix of the sample firms over time, we require each firm to have non-missing observations in each of the 6 quarters over our sample period. This sample restriction results in a loss of 740 firm quarters in column (1) and 641 firm quarters in column (2). As shown in column (1), the coefficient on AFTER is significantly positive, which is inconsistent with the alternative hypothesis that controlling shareholders' expropriation of minority shareholders using inter-

¹⁹ Inference is similar if OREC is defined using other receivables net of the allowance for bad debt expense. We prefer to use gross other receivables because the reporting for the allowance for bad debt expense is subject to considerable managerial discretion.

corporate loans declined around the 2004 regulation. In addition, the coefficients between AFTER and the three minority shareholder ownership structure variables are insignificant. Overall, these results reduce the concern that our results in Tables 2 and 3 are due to a general improvement in investor protection over our sample period.²⁰ Our results are consistent with Jiang et al. (2010) who find that the inter-corporate loan problem was not resolved until November 7, 2006 when eight Chinese government ministries issued a joint announcement that would hold the top management of controlling shareholders personally accountable for failing to pay back inter-corporate loans to A share firms.²¹

Third, we examine whether the decrease in value decreasing equity offering proposal announcements in the post-regulation period shown in Tables 2 and 3 is due to management's anticipation of the split share structure reform. The CSRC announced the first pilot batch of four companies for the reform in early May 2005, but the reform was expanded to all listed firms by August 2005 (Li et al. 2008). The expected launch of the reform has two potential effects on management's incentive to submit equity offering proposals. The first effect of the reform is that because the CSRC is expected to cease processing equity offering proposals upon the launch of the reform, management may have a lower incentive to announce both value increasing and value decreasing equity offering proposals in the post-regulation period. This prediction is not

²⁰ Inference is similar if the sample in Table 5 is limited to firms whose OREC at the 2003 year-end is above the median or only the firms included in Table 2.

²¹ Jiang et al. (2010) also find that the level of OREC is negatively associated with the listed firm's future earnings performance. This negative relation holds in our sample as well. In addition, the negative relation is similar for both the pre- and post-regulation periods, suggesting that the nature of OREC is similar over the two time periods.

consistent with Table 2 because we find different results for value increasing and value decreasing proposals. In addition, our results in Table 2 are robust to excluding the 2nd quarter of 2005, which is more likely subject to the anticipation effect of the reform.

The second effect of the split share structure reform is that the increased liquidity of non-tradable shares post the reform should help align the interests between controlling shareholders (especially those with a larger ownership of non-tradable shares) and minority shareholders and therefore controlling shareholders of A share firms may have a weaker incentive to expropriate minority shareholders (e.g., announcing value decreasing equity offering proposals) even in our post-regulation period, which predates the completion of the reform. To rule out this alternative explanation, we rerun the regressions in Tables 2 and 3 after including NONTRADE_OWN (defined as the stock ownership of all non-tradable shareholders) and its interaction with AFTER. To the extent that the documented results in Tables 2 and 3 are due to this alternative explanation, the marginal effect of AFTER*NONTRADE_OWN should be significantly negative for value decreasing proposals in Table 2 and significantly positive in Table 3. In addition, including AFTER*NONTRADE_OWN would reduce the marginal effect of AFTER*MUTUAL_OWN in both Tables 2 and 3. Untabulated results show that the mean marginal effect of AFTER*NONTRADE_OWN is never significantly different from zero in all the regressions, suggesting no evidence of an anticipation effect of the reform. More importantly, the mean marginal effect of AFTER*MUTUAL_OWN for value decreasing proposals remains significantly negative for the regression in Table 2 and significantly positive for the regression in

Table 3, suggesting that our results are not driven by management's anticipation of the split share restructure reform.

4. Proposal quality and minority shareholders' voting behavior in the post-regulation period

Conditional on management submitting an equity offering proposal for shareholder approval, we use the following logit model to examine whether minority shareholders are more likely to veto (pass) management's value decreasing (increasing) equity offerings proposals in the post-regulation period:

$$VETO_{it} = a + b * DCAR_{it} + \varepsilon_{it} \quad (3)$$

where i and t are proposal and date indicators, respectively. $VETO$ is a dummy variable that equals 1 if a proposal is vetoed by minority shareholders, and zero if it is passed by minority shareholders. $DCAR$ is a dummy variable that is equal to one if for value increasing equity offering proposals (i.e., $CAR > 0$) and zero otherwise. Regression model (3) is different from regression model (1) in that we do not compare the regression coefficients across the pre- and post-regulation periods. As management had an absolute say on the equity offering decision in the pre-regulation period, any proposals submitted by management would be passed without exception. Therefore, it is not meaningful to run regression model (3) in the pre-regulation period (i.e., the coefficient on $DCAR$ should be always zero).

We prefer to use DCAR instead of CAR as a proxy for proposal quality for two reasons. First, the relation between CAR and VETO is unlikely to be a linear function. While minority shareholders should care about whether a proposal is value increasing or value decreasing (i.e., the sign of CAR), they should be less concerned about the magnitude of proposal quality (i.e., the magnitude of CAR). For example, two value decreasing proposals' CARs could differ significantly (e.g., -10% versus -20%), but we expect minority shareholders to veto both with equal likelihood. Thus, we believe that using DCAR is more appropriate for regression model (3). Second, CAR is subject to greater endogeneity than DCAR due to the stock price's anticipation of minority shareholders' voting outcomes in the post-regulation period. Brickley et al. (1988) argue that DCAR can be used as an exogenous instrument for the endogenous CAR. While the magnitude of CAR could be affected by the stock market's anticipation of the likelihood of minority shareholders' voting outcome, it is likely that CAR is still negative (positive) for value decreasing (increasing) proposals considering the fact that there is still some uncertainty on the eventual voting outcome by minority shareholders. Thus, DCAR should have the ability to separate value increasing ($CAR > 0$) proposals from value decreasing ($CAR < 0$) proposals and thus could serve as a valid (though may not be a perfect) exogenous instrument for proposal quality.²²

The coefficient on DCAR is difficult to predict. Anticipating minority shareholders' vetoing of value reducing proposals, in theory rational managers should not submit value

²² As expected, the marginal effects of DCAR and $DCAR * MUTUAL_OWN$ in models 1 and 2 of Table 6 respectively are insignificant if DCAR is replaced with CAR.

reducing equity offering proposals any more in the post-regulation period. Therefore, we should expect the coefficient on DCAR to be insignificant.²³ However, if some managers continue to submit value decreasing proposals for unknown reasons, we should expect rational minority shareholders to veto such proposals and therefore the coefficient on DCAR should be significantly negative.

We also examine whether a firm's top 10 minority shareholder composition affects the effect of DCAR on VETO. Specifically, we allow the coefficient on DCAR to vary with *MUTUAL_OW**N*, *OTHERINST_OW**N* and *INDIVIDUAL_OW**N*, all of which are measured at the fiscal quarter end immediately before the proposal voting date. Top 10 minority shareholders can affect a proposal's voting outcome directly by casting their own votes in a certain way or indirectly by influencing other shareholders' voting behavior.

Model 1 in Panel A of Table 6 shows the regression result of model (3). There are 82 equity offering proposals that were voted on by minority shareholders in the post-regulation period. We exclude three proposals withdrawn prior to the minority shareholder voting date. Results are similar if we treat the withdrawn proposals as vetoed proposals. Though the mean and median CAR for the 82 proposals is significantly positive, CAR is negative for 35 out of the 82 proposals. 10 out of the 82 proposals (12%) were vetoed by minority shareholders (untabulated), suggesting that minority shareholders did exercise the new control power granted

²³ However, it is important to note that an insignificant coefficient on DCAR does not imply that the 2004 regulation is ineffective because the regulation has a significant deterrence effect on value decreasing proposals as shown in Tables 2 and 3.

by the 2004 regulation. The coefficient on DCAR in Panel A of Table 6 is negative but insignificant. Thus, conditional on the proposals submitted by management, on average there is no evidence that value decreasing (value increasing) proposals are more likely to be vetoed (passed) by voting minority shareholders.

Model 2 in Panel A of Table 6 reports the logit regression coefficients of model (3) that allows the coefficient on DCAR to vary with MUTUAL_OWN, OTHERINST_OWN, and INDIVIDUAL_OWN. As this is a logit model, we also report the correct mean marginal effects and associated mean z-statistics for the three interaction variables in Panel B of Table 6. Appendix B shows the formula for the marginal effect of an interaction term in a logit model with multiple interaction variables. The mean marginal effect of DCAR*MUTUAL_OWN is significantly negative at the 10% two-tailed level but the mean marginal effects of DCAR*OTHERINST_OWN and DCAR*INDIVIDUAL_OWN are insignificant. Overall, the results in Table 6 suggest that minority shareholders are more likely to veto value decreasing proposals in firms with higher mutual fund ownership, a finding consistent with those in Tables 2 and 3.

5. The stock market reaction to the announcement of the 2004 regulation

This section conducts an alternative test on the effect of the 2004 regulation on management's incentive to expropriate minority shareholders by examining the Chinese domestic stock market's reaction to the announcement of the 2004 regulation. While the analyses

in Sections 3 and 4 use equity offering proposals as a specific example to directly demonstrate the effect of the 2004 regulation on the quality of corporate decisions, the advantage of the stock market reaction test is that it can capture all the expected effects of the regulation, both direct (including equity offering proposals) and indirect (e.g., changes in management's tunneling behavior as a result of the 2004 regulation).

5.1. Methodology

We identify two key events relevant to the 2004 regulation (denoted events 1 and 2, respectively). Event 1 is the public release of the exposure draft of the regulation on September 27, 2004 and event 2 is the approval of the final regulation on December 7, 2004. We use a 5-trading day window centered on the two event dates (i.e., $[-2, +2]$) to measure the market reaction to both events, though inferences are similar if we use a longer return holding period $[-2, +10]$ around an event (the same holding period as the proposal quality proxy CAR in Section 3) to account for the potential delay in the market reaction to an event (untabulated).²⁴

We first use the following time series regression model to assess the overall stock market reactions to the two events associated with the regulation over a 250-trading day period ending on December 10, 2004 (i.e., the last trading day of event 2):

²⁴ We find no evidence of any price run-up over the $[-7, -3]$ trading days prior to the two events.

$$R_{mt} = a + \sum_{i=-1}^{i=+1} (b_i * R_{hkm,t+i}) + c * EVENT1_t + d * EVENT2_t + \varepsilon_t \quad (4)$$

R_{mt} is the equally weighted dividend inclusive market return of all A share firms on trading day t.

$R_{hkm,t}$ is the equally weighted dividend inclusive market return of all H share firms and Red Chip share firms that are not listed on the two domestic stock exchanges.²⁵ H shares are defined as mainland Chinese investor controlled firms that are incorporated in mainland but listed in Hong Kong and Red Chip firms are mainland Chinese investor controlled firms that are incorporated outside China and traded in Hong Kong. $R_{hkm,t}$ is set to zero if the Hong Kong Stock Exchange is closed on trading day t. $R_{hkm,t}$ controls for the market return unrelated to the 2004 regulation.

The H shares and Red Chip shares included in $R_{hkm,t}$ are not subject to the 2004 regulation. In addition, as noted in Ke et al. (2010), most H shares and Red Chip shares operate their main businesses in mainland China and thus should be subject to the same economic forces as domestically listed A share firms. Thus, $R_{hkm,t}$ should be a reasonable control for the confounding effects unrelated to the regulation. However, we also conduct a sensitivity check in Section 5.2 by retaining only the A share firms that are most comparable to the H/Red Chip firms included in the computation of $R_{hkm,t}$.

$EVENT1$ is a dummy variable that is equal to 0.2 for a trading day that falls within the [-2, +2] event window centered on event 1, and zero otherwise. $EVENT2$ is a dummy variable that

²⁵ Results are similar if we use value weighted market returns.

is equal to 0.2 for a trading day that falls within the $[-2, +2]$ event window centered on event 2, and zero otherwise. Because of the way we define *EVENT1* and *EVENT2*, the coefficients on *EVENT1* and *EVENT2* represent the cumulative abnormal return over the 5 trading days centered on event 1 and event 2, respectively.

As noted in Section 3, the efficacy of the 2004 regulation hinges on whether and how minority shareholders vote on submitted managerial proposals. Hence, we also examine how the stock market reactions to events 1 and 2 vary with *MUTUAL_OWN*, *OTHERINST_OWN*, and *INDIVIDUAL_OWN* using the Sefcik and Thompson (1986) methodology, which controls for the cross-sectional dependence and heteroscedasticity in contemporaneous stock returns. The three minority shareholder ownership variables are defined as in Section 3 except that they are all measured at the end of the quarter immediately before a relevant event. To control for the common firm size and book-to-market effects, we also include $\ln(MV)$ and $\ln(BM)$ as control variables. *MV* is defined as the product of stock price and total common shares outstanding, measured at the end of the quarter immediately before an event. *BM* is defined as the book value of common equity divided by *MV*, both measured at the end of the quarter immediately before an event.²⁶ Following Fama and French (1992), we take the natural logarithm of *MV* and *BM* to reduce the skewness of both variables.

²⁶ Since non-tradable shares are often sold at a huge discount relative to tradable shares, we also define *MV* as the sum of the book value of non-tradable shares and the market value of tradable shares and obtain similar inferences.

5.2. Results

Panel A of Table 7 shows the regression result of model (4). The coefficients on $R_{hkm,t-1}$ and $R_{hkm,t+1}$ are insignificant, but the coefficient on $R_{hkm,t}$ is significantly positive, suggesting that common contemporaneous economic forces affect the returns of both A share firms and H/Red Chip shares. The insignificant coefficients on EVENT1 and EVENT2 suggest that there is no evidence that investors expect the regulation to have a significant impact on shareholder value for the overall A share stock market. This finding is inconsistent with the significant deterrence effect of the 2004 regulation shown in Panel A of Table 2 and Panel B of Table 3.

There are two non-mutually exclusive explanations for this inconsistency in results. First, the 2004 regulation has indirect negative effects on shareholder value that offset the positive effects of the regulation. For example, the regulation may force management to take alternative tunneling activities that could be more costly to minority shareholders. Second, the event study results in Panel A of Table 7 are influenced by uncontrolled-for confounding news events.

To determine whether confounding news events are responsible for the insignificant abnormal returns in Panel A, Panel B of Table 7 shows the coefficients on the interaction terms between the two event dummies and the three top 10 minority shareholder ownership characteristics. Relative to the results in Panel A, Panel B's results are immune from uncontrolled-for confounding news events that affect all A share firms because the interaction terms in Panel B capture the differences in event period abnormal returns across firms with different ownership structures. The coefficient on EVENT1*INDIVIDUAL_OWNS is

insignificant, suggesting that individual investor ownership does not have a significant impact on the stock market's reaction to the regulation. The coefficients on $EVENT1 * MUTUAL_OWN$ and $EVENT1 * OTHERINST_OWN$ are both significantly positive but we will show in Panel C of Table 7 that the significant coefficient on $EVENT1 * OTHERINST_OWN$ is not robust. In addition, the coefficient on $EVENT1 * MUTUAL_OWN$ is larger than the coefficient on $EVENT1 * OTHERINST_OWN$ (two-tailed $p < 0.001$). None of the coefficients between $EVENT2$ and the ownership variables is significant.

Ke et al. (2010) find that A share firms are typically much smaller than H/Red Chip firms, which are usually large monopolies in certain industries. Hence, $R_{hkm,t}$ may not be an adequate control for the market return unrelated to the 2004 regulation for smaller A share firms that operate in different industries. To reduce this concern, we redo the regression results in Table 7 for only the A share firms that are in the same industry (Datastream INDC2) as the H/Red Chip firms and whose total assets at the end of fiscal year 2003 fall between 80% and 120% of the median total assets of the H/Red Chip firms in the same industry. Results are similar if we choose tighter cutoffs of 90% and 110%. These sample restrictions result in a final sample of 187 A share firms. As shown in Panel C of Table 7, the coefficient on $EVENT1 * MUTUAL_OWN$ continues to be significantly positive. Moreover, the coefficient on $EVENT2 * MUTUAL_OWN$ becomes significantly positive, too. However, the coefficients on $EVENT1 * OTHERINST_OWN$ and $EVENT1 * INDIVIDUAL_OWN$ are not significant at the 10% significance level. We conclude from Panel C that the stock market reaction to the regulation is more positive for firms

with higher mutual fund ownership, but not for firms with other types of top 10 minority shareholder ownership. Overall, the results in Table 7 are consistent with the equity offering results in Tables 2, 3, and 6.

5.3. Confounding events

One potential concern with our event study is that the results in Table 7 could be due to other regulatory announcements issued at the same time as the segmented voting rule. To rule out this alternative explanation, we identified all the regulatory announcements issued by the CSRC, the two Chinese stock exchanges, and the State Council during our two event windows. There are 10 regulatory announcements during the first event window and 7 regulatory announcements during the 2nd event window. However, all of the confounding regulatory announcements deal with procedural and administrative issues and thus are unlikely to have a material impact on stock prices.²⁷

We also examine whether the results in Table 7 can be explained by the other minor investor protection provisions included in the 2004 regulation (see footnote 17 for a summary of the four provisions). While a portion of the documented stock market reactions in Table 7 could be due to these additional provisions, we believe that most of the stock market reactions should be attributed to the segmented voting provision for the following reasons. First, both the CSRC

²⁷ For example, both the Shanghai and Shenzhen stock exchanges issued a routine notice on September 27, 2004 urging all listed firms to do a good job in preparing for the third quarter financial reports. The list of the 17 government regulations is available upon request.

and media reports predominantly focused on the segmented voting rule, suggesting that the segmented voting rule was the most important provision of the 2004 regulation (see, e.g., Peng 2004). Second, unlike the segmented voting provision, the other provisions are difficult to enforce by either regulators or minority shareholders and therefore are unlikely to be viewed as equally effective by minority shareholders. For example, due to concentrated ownership and weak investor protections in mainland China, it is well recognized that so-called independent directors in provision (a) are often captured by management. Provision (b) is not even mandatory. Management can easily circumvent provision (c) by paying a nominal amount of dividends before issuing new equity. There is no compelling reason to believe that provision (d) is effective because the CSRC has never strictly enforced its own regulations. Finally, none of the above mentioned confounding provisions would predict a differential stock market reaction for firms with higher mutual fund ownership. This is because except for the control power granted by the segmented voting provision, A share firms' minority shareholders are powerless in corporate decision making due to the dominance of controlling shareholders.

6. Conclusion

An important issue hotly debated among investors and regulators around the world is the allocation of control over corporate decisions between management/controlling shareholders (referred to as management for brevity) and minority shareholders. We use a unique Chinese securities regulation to directly test the effect of granting minority shareholders direct control

over corporate decisions on shareholder value. In December 2004, the Chinese securities regulator, China Securities Regulatory Commission (CSRC), issued a new regulation that requires several types of major corporate decisions to seek the separate approval of tradable shareholders (referred to as minority shareholders). Equity offering is the most frequent proposal subject to the regulation.

Our primary findings can be summarized as follows. First, the regulation helps deter management from submitting value decreasing equity offering proposals, especially for firms with higher mutual fund ownership. In addition, the stock market reaction to the announcement of equity offering proposals is less negative in the post-regulation period, especially for firms with higher mutual fund ownership. Second, there is evidence that value reducing equity offering proposals submitted in the post-regulation period are more likely to be vetoed by minority shareholders in firms with higher mutual fund ownership. Third, the overall stock market reaction to the announcement of the 2004 regulation is insignificant, but the stock market reaction is more positive for firms with higher mutual fund ownership. Overall, our results suggest that the 2004 regulation increases shareholder value in firms with higher mutual fund ownership.

Our study provides valuable information to the debate on the costs and benefits of granting minority shareholders direct control over corporate decisions. Our results are directly relevant to the Chinese securities regulator (CSRC) who faces a daunting task of protecting minority shareholders' interests and developing the country's domestic financial market. We

provide direct evidence on the overall effect of a regulation that strengthens minority shareholders' direct control over corporate decisions on shareholder value. In addition, we identify the roles of minority shareholder composition (especially the role of mutual funds) in determining the effectiveness of such regulations. Our results are also of interest to other countries' regulators who are considering legislations that would grant minority shareholders more direct control over corporate decisions.

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Appendix A. The procedures used to identify the firms eligible to propose equity offerings

We rely on the following regulations issued by the Chinese Securities Regulatory Commission (CSRC) to identify the firms that are eligible to propose an equity offering (a general offering, a rights offering, or a convertible bond offering): a) Measures for the Administration of New Share Issuance by Listed Companies (Order No. 1 [2001] of the CSRC); b) Notice on the Administration of New Share Issuance by Listed Companies (Order No. 43 [2001] of the CSRC); c) Implementation Measures for Listed Companies' Issuing Convertible Corporate Bonds (Order No. 2 [2001] of the CSRC); d) The Interim Measures for the Administration of Convertible Corporate Bonds (Order No. 16 [1997] of the Securities Committee of the State Council; e) Notice on the Administration of Convertible Corporate Bond Issuance by Listed Companies (Order No. 115 [2001] of the CSRC); f) Notice on the Conditions for the Additional Issuance of Securities by Listed Companies (Order No. 2 [2002] of the CSRC); and g) Notice of the China Securities Regulatory Commission on Several Issues Concerning Major Purchases, Sales and Exchanges of Assets by Listed Companies (Order No. 105 [2001] of the CSRC). Under these regulations firms that wish to propose an equity offering are required to meet several qualitative and quantitative requirements. Though the quantitative requirements are generally straightforward, most qualitative requirements are subjective and difficult to measure using publicly available data. Hence, we rely on the quantitative requirements to determine a firm's equity offering eligibility. Specifically, a firm is deemed eligible to propose a *rights offering* if it satisfies the following two conditions: a) the average return on equity (ROE) over the past three years is no less than 6%; and b) the firm has not conducted any rights offering in the previous year. A firm is deemed eligible to propose a *general offering* if it satisfies the following two conditions: a) the average ROE (based on an unknown formula specified by the CSRC) over the past three years is no less than 10%;²⁸ and b) the ROE in the previous year is no less than 10%. However, under the CSRC regulations a firm is also deemed eligible to propose a general offering if it experiences a "significant" restructuring in any of the previous three years. A restructuring is deemed significant if the restructuring's deal value is no less than 50% of the firm's gross total assets. A firm is deemed eligible to propose a *convertible bond offering* if it

²⁸ As we do not have access to the CSRC's ROE formula, we define ROE as annual net income divided by the average shareholder's equity.

satisfies the following two conditions: a) the average ROE over the past three years is no less than 10% or the average ROE based on net income excluding non-recurring items over the past three years is no less than 6%; and b) the firm does not report a loss in any of the previous three years. Again, under the CSRC regulations a firm is also deemed eligible to propose a convertible bond offering if it experiences a “significant” restructuring in any of the previous three years.

A firm year is excluded from our sample of eligible firms if it does not satisfy the eligibility requirements for a general offering, a rights offering, or a convertible bond offering. If we literally follow the above eligibility requirements, a significant number of firm years that did propose equity offerings would be excluded. As a result, we relax the quantitative thresholds by reducing the 10% threshold to 9%, the 6% threshold to 5%, and the 50% threshold to 40%. With those relaxed thresholds, all but one equity offering proposals are retained in our final sample.

Appendix B. The correct marginal effects of interaction variables in nonlinear regressions

A. The marginal effects of interaction terms for the multinomial logit model in Table 2

The multinomial logit model in Panel B of Table 2 is as follows:

$$\Phi_J \equiv \Pr[Y = J] = \frac{e^{Z_J}}{1 + \sum_{j=1}^2 e^{Z_j}} \quad (A)$$

where $J=1$ for value increasing proposals and 2 for value decreasing proposals, and

$$\begin{aligned} Z_J \equiv & a_J + b_{J,1}MUTUAL_OWN + b_{J,2}OTHERINST_OWN + b_{J,3}INDIVIDUAL_OWN + \\ & c_J AFTER + \\ & d_{J,1}AFTER * MUTUAL_OWN + d_{J,2}AFTER * OTHERINST_OWN + \\ & d_{J,3}AFTER * INDIVIDUAL_OWN + \gamma_J CONTROL + e_J \end{aligned}$$

Denote $\Phi_J[0] = \Phi_J[AFTER = 0]$ and $\Phi_J[1] = \Phi_J[AFTER = 1]$

The three marginal effects of $AFTER * MUTUAL_OWN$, $AFTER * OTHERINST_OWN$, and $AFTER * INDIVIDUAL_OWN$ in the multinomial logit model are defined as follows:

$$\begin{aligned} & \frac{\partial[\Phi_J[1] - \Phi_J[0]]}{\partial MUTUAL_OWN} \\ &= \Phi_J[1] \left[(b_{J,1} + d_{J,1}) - \sum_{j=1}^2 [\Phi_j[1] \cdot (b_{j,1} + d_{j,1})] \right] - \Phi_J[0] \cdot \left[b_{J,1} - \sum_{j=1}^2 \Phi_j[0] b_{j,1} \right] \end{aligned} \quad (A1)$$

$$\begin{aligned} & \frac{\partial[\Phi_J[1] - \Phi_J[0]]}{\partial OTHERINST_OWN} \\ &= \Phi_J[1] \left[(b_{J,2} + d_{J,2}) - \sum_{j=1}^2 [\Phi_j[1] \cdot (b_{j,2} + d_{j,2})] \right] - \Phi_J[0] \cdot \left[b_{J,2} - \sum_{j=1}^2 \Phi_j[0] b_{j,2} \right] \end{aligned} \quad (A2)$$

$$\begin{aligned} & \frac{\partial[\Phi_J[1] - \Phi_J[0]]}{\partial INDIVIDUAL_OWN} \\ &= \Phi_J[1] \left[(b_{J,3} + d_{J,3}) - \sum_{j=1}^2 [\Phi_j[1] \cdot (b_{j,3} + d_{j,3})] \right] - \Phi_J[0] \cdot \left[b_{J,3} - \sum_{j=1}^2 \Phi_j[0] b_{j,3} \right] \end{aligned} \quad (A3)$$

B. The marginal effects of interaction terms for the Logit model in Table 6 (a special case of the multinomial Logit model)

The logit model in Panel B of Table 6 is as follows

$$\Phi \equiv \Pr[VETOED] = \frac{e^Z}{1 + e^Z} \quad (B)$$

where

$$Z = a + b_1 MUTUAL_OWN + b_2 OTHERINST_OWN + b_3 INDIVIDUAL_OWN + c \cdot DCAR + d_1 DCAR * MUTUAL_OWN + d_2 DCAR * OTHERINST_OWN + d_3 DCAR * INDIVIDUAL_OWN + e$$

Denote $\Phi[0] = \Phi[DCAR = 0]$ and $\Phi[1] = \Phi[DCAR = 1]$.

The three marginal effects of $DCAR * MUTUAL_OWN$, $DCAR * OTHERINST_OWN$, and $DCAR * INDIVIDUAL_OWN$ in the logit model are defined as follows:

$$\frac{\partial[\Phi[1] - \Phi[0]]}{\partial MUTUAL_OWN} = \Phi'[1] \cdot [b_1 + d_1] - \Phi'[0] \cdot b_1 \quad (B1)$$

$$\frac{\partial[\Phi[1] - \Phi[0]]}{\partial OTHERINST_OWN} = \Phi'[1] \cdot [b_2 + d_2] - \Phi'[0] \cdot b_2 \quad (B2)$$

$$\frac{\partial[\Phi[1] - \Phi[0]]}{\partial INDIVIDUAL_OWN} = \Phi'[1] \cdot [b_3 + d_3] - \Phi'[0] \cdot b_3 \quad (B3)$$

Table 1. Descriptive Statistics (N=3,999 firm-quarter observations)

CAR is the market adjusted cumulative abnormal return over the [-2, +10] trading days around the equity offering proposal announcement date. SUBMISSION = 0 if a firm does not submit a proposal in quarter t, 1 if a firm submits a value increasing (i.e., $CAR > 0$) proposal in quarter t, and 2 if a firm submits a value decreasing (i.e., $CAR \leq 0$) proposal in quarter t. MUTUAL_OWNS is the total stock ownership (as a percentage of the total outstanding tradable shares) of all the open ended and close ended mutual funds ranked among the top 10 minority shareholders at the end of the quarter immediately before the proposal date. OTHERINST_OWNS is the total stock ownership (as a percentage of the total outstanding tradable shares) of all the other institutional investors ranked among the top 10 minority shareholders at the end of the quarter immediately before the proposal date. INDIVIDUAL_OWNS is the total stock ownership (as a percentage of the total outstanding tradable shares) of all the individual investors ranked among the top 10 minority shareholders at the end of the quarter immediately before the proposal date. Q is the natural logarithm of a firm's Tobin's Q at the beginning of quarter t. Q is defined as the market value minus the book value of shareholders' equity plus total assets divided by total assets. Results are similar if the market value of non-tradable shares is assumed equal to their book value in the Q definition. CFO is cash flows from operations over quarters t-4 to t-1 divided by the average total assets at the beginning of quarter t. LEV is total debts divided by total assets at the beginning of quarter t. CASH is cash and marketable securities divided by total assets at the beginning of quarter t. VOLATILITY is the standard deviation of daily stock returns over a one year period that ends in the beginning of quarter t. AR12 is the buy and hold equally weighted market adjusted abnormal return over a one-year period that ends. ASSETS is the natural logarithm of total assets at the beginning of quarter t.

Variable name	Mean	Std	Min	Q1	Median	Q3	Max
SUBMISSION=1	0.026	0.159	0.000	0.000	0.000	0.000	1.000
SUBMISSION=2	0.030	0.170	0.000	0.000	0.000	0.000	1.000
MUTUAL_OWNS	0.046	0.081	0.000	0.000	0.004	0.055	0.851
OTHERINST_OWNS	0.063	0.134	0.000	0.003	0.018	0.063	0.996
INDIVIDUAL_OWNS	0.020	0.020	0.000	0.004	0.017	0.030	0.179
Q	0.576	0.359	-0.135	0.329	0.514	0.762	6.435
CASH	0.172	0.119	0.002	0.084	0.143	0.229	0.584
CFO	0.071	0.107	-0.354	0.017	0.072	0.127	0.426
LEV	0.465	0.188	0.056	0.339	0.466	0.588	2.685
VOLATILITY	0.023	0.011	0.011	0.018	0.021	0.024	0.111
AR12	0.061	0.284	-0.573	-0.124	-0.003	0.190	1.212
ASSETS	21.424	1.084	14.686	20.729	21.309	22.011	27.169

Table 2. The effect of the 2004 regulation on management's incentive to submit value increasing and value decreasing equity offering proposals (N=3,999 firm quarter observations)

AFTER is a dummy variable that is equal to one for the quarters in the post-regulation period (i.e., after December 7, 2004), and zero for the quarters in the pre-regulation period. AFTER is coded zero for the proposals submitted prior to December 7 in the 4th calendar quarter of 2004. The proposals submitted on or after December 7 in the 4th quarter of 2004 are treated as proposals submitted in the 1st quarter of 2005. As the number of months for calendar quarter 4 of 2004 is 2 while the number of months for calendar quarter 1 of 2005 is 4, we include the number of months in each calendar quarter as a control in estimating model (2). See Table 1 for other variable definitions. Two-tailed robust p values clustered at the firm level are reported in parentheses for Panels A and B. The mean marginal effects and z-statistics are computed using the formulas in Appendix B.

Panel A. Main effects model

	SUBMISSION=1 (value increasing)	SUBMISSION=2 (value decreasing)
AFTER	-0.1923 (0.526)	-1.1233 (0.000)
LEV	1.3979 (0.001)	0.7993 (0.059)
Q	-0.8163 (0.060)	-0.4207 (0.238)
CFO	0.8927 (0.373)	1.9722 (0.052)
VOLATILITY	-14.7321 (0.281)	-26.9777 (0.007)
AR12	1.4978 (0.000)	0.7135 (0.020)
CASH	-2.8388 (0.006)	-3.9386 (0.000)
ASSETS	-0.1647 (0.268)	-0.2149 (0.047)
MONTHS	0.8543 (0.001)	1.1525 (0.000)
CONSTANT	-2.3436 (0.477)	-1.2180 (0.630)
Pseudo R-square	0.0517	

Panel B. Interaction effects model

	SUBMISSION=1 (value increasing)	SUBMISSION=2 (value decreasing)
MUTUAL_OW	0.8968 (0.642)	1.9890 (0.197)
OTHERINST_OW	0.2950 (0.785)	-0.2725 (0.759)
INDIVIDUAL_OW	6.4982 (0.324)	9.6482 (0.094)
AFTER	0.1108 (0.805)	0.1657 (0.771)
AFTER*MUTUAL_OW	1.0212 (0.658)	-9.3129 (0.007)
AFTER*OTHERINST_OW	-1.1001 (0.512)	-1.7289 (0.436)
AFTER*INDIVIDUAL_OW	-22.9151 (0.075)	-40.2507 (0.031)
LEV	1.5063 (0.001)	0.7913 (0.089)
Q	-0.9547 (0.039)	-0.4181 (0.239)
CFO	0.7090 (0.476)	2.1188 (0.037)
VOLATILITY	-12.5339 (0.343)	-26.4886 (0.010)
AR12	1.3752 (0.000)	0.8103 (0.025)
CASH	-2.7946 (0.009)	-3.8674 (0.000)
ASSETS	-0.2183 (0.233)	-0.1979 (0.135)
MONTHS	0.8647 (0.001)	1.1705 (0.000)
CONSTANT	-1.3989 (0.729)	-1.9484 (0.528)
Pseudo R-square	0.0616	

Panel C. Mean marginal effect (mean Z-statistic)

	SUBMISSION=1 (value increasing)	SUBMISSION=2 (value decreasing)
mean marginal effect of AFTER*MUTUAL_OW	0.0261 (0.337)	-0.2016 (-1.833)
mean marginal effect of AFTER*OTHERINST_OW	-0.0247 (-0.589)	-0.0196 (-0.339)
mean marginal effect of AFTER*INDIVIDUAL_OW	-0.4917 (-1.415)	-0.8796 (-1.905)

Table 3. The effect of the 2004 regulation on equity offering proposal quality

See Tables 1 and 2 for variable definitions. Two-tailed robust p values shown in Panel C are clustered at the firm level.

Panel A. Differences in firm characteristics for the submitted equity offering proposals in the pre- and post-regulation periods

Variable	Pre-regulation			Post-regulation			p-value on the test of difference	
	N	Mean	Median	N	Mean	Median	t-test	Rank-sum test
Q	147	0.593	0.535	81	0.453	0.372	(0.002)	(0.001)
CASH	147	0.144	0.131	81	0.123	0.107	(0.079)	(0.084)
LEV	147	0.503	0.501	81	0.544	0.567	(0.060)	(0.036)
CFO	144	0.080	0.081	80	0.086	0.087	(0.655)	(0.582)
VOLATILITY	147	0.020	0.019	81	0.025	0.022	(0.004)	(0.000)
AR120	147	0.115	0.041	81	0.121	0.054	(0.884)	(0.663)
ASSETS	147	21.626	21.370	81	21.611	21.488	(0.923)	(0.601)

Panel B. The market reactions to announcements of equity offering proposals in the pre- and post- regulation periods: mean (median) [S.D.]

	CAR in the pre-regulation period	CAR in the post-regulation period	Two-tailed p value on the test of the difference	
			t-test	rank-sum test
	-0.014 (-0.018) [0.068]	0.014 (0.018) [0.066]	0.003	0.004
Two-tailed p value of one-sample t-test	0.012	0.069		
Two-tailed p value of one-sample rank-sum test	0.005	0.119		

Panel C. OLS regression result of CAR: interaction effects model (N=228)

	Coefficient (two-tailed p)
CONSTANT	-0.0147 (0.121)
MUTUAL_OWN	-0.0261 (0.758)
OTHERINST_OWN	0.0099 (0.760)
INDIVIDUAL_OWN	0.0475 (0.796)
AFTER	-0.0063 (0.719)
AFTER*MUTUAL_OWN	0.3749 (0.002)
AFTER*OTHERINST_OWN	0.0044 (0.938)
AFTER*INDIVIDUAL_OWN	0.4714 (0.443)
Adjusted R-square	0.071

Table 4. Replication of the interaction effects models shown Panel C of Tables 2 and 3 using three alternative 6-quarter time periods

The three alternative time periods are (a) 2003Q1-2004Q2; (b) 2003Q2-2004Q3; and (c) 2003Q3-2004Q4, respectively. Similar to the definition of *AFTER*, the pseudo *AFTER* is one for the last two quarters and zero for the first four quarters for each of the three 6-quarter time periods. See Tables 2 and 3 for other variable definitions.

Panel A. Replication of Panel C of Table 2 using pseudo *AFTER*: mean marginal effect (mean Z-statistic)

	2003Q1-2004Q2		2003Q2-2004Q3		2003Q3-2004Q4	
	SUBMISSION=1 (value increasing)	SUBMISSION=2 (value decreasing)	SUBMISSION=1 (value increasing)	SUBMISSION=2 (value decreasing)	SUBMISSION=1 (value increasing)	SUBMISSION=2 (value decreasing)
AFTER*MUTUAL_OWN	0.0421 (0.523)	0.0746 (0.776)	0.0212 (0.380)	0.0223 (0.347)	-0.0656 (-0.861)	0.0511 (0.699)
AFTER*OTHERINST_OWN	0.0173 (0.434)	0.0001 (0.006)	0.0061 (0.174)	0.0927 (2.065)	0.0054 (0.141)	0.0038 (0.135)
AFTER*INDIVIDUAL_OWN	0.3022 (1.709)	0.4386 (1.469)	0.2067 (0.822)	-0.0329 (-0.176)	-0.4736 (-1.045)	0.1923 (0.606)

Panel B. Replication of Panel C of Table 3 using pseudo *AFTER*: the coefficients for the key variables of interest

	Coefficient (two-tailed p)		
	2003Q1-2004Q2	2003Q2-2004Q3	2003Q3-2004Q4
AFTER*MUTUAL_OWN	-0.0220 (0.901)	0.0562 (0.755)	-0.2023 (0.225)
AFTER*OTHERINST_OWN	-0.0108 (0.872)	-0.1356 (0.034)	-0.0778 (0.247)
AFTER*INDIVIDUAL_OWN	0.5045 (0.192)	-0.1239 (0.817)	-1.2365 (0.003)
Adjusted R-square	-0.015	-0.019	-0.005

Table 5. The regression results of inter-corporate loans

OREC is gross other receivables divided by year-end total assets. LNTA is the natural logarithm of year-end total assets. All variables are winsorized at the 1% and 99% percentiles. See Table 1 for other variable definitions. MUTUAL_OWN, OTHERINST_OWN, and INDIVIDUAL_OWN are measured at the beginning of the quarter. The sample in each column includes all A share firms that have nonmissing data in each of the 6 quarters over 1/1/2004-6/30/2005. Two-tailed robust p values clustered at the firm level are reported in parentheses.

	Dependent variable = OREC	
	(1)	(2)
AFTER	0.0076 (0.000)	0.0059 (0.035)
MUTUAL_OWN		0.0102 (0.409)
OTHERINST_OWN		-0.0200 (0.127)
INDIVIDUAL_OWN		-0.1508 (0.065)
After*MUTUAL_OWN		-0.0088 (0.504)
AFTER*OTHERINST_OWN		-0.0005 (0.960)
AFTER*INDIVIDUAL_OWN		0.0765 (0.412)
LNTA	-0.0466 (0.005)	-0.0471 (0.007)
Firm fixed effects	YES	YES
Adj. R ²	0.89	0.89
N	6,906	6,879

Table 6. Proposal quality and the likelihood of minority shareholders' veto in the post-regulation period (N=82 proposals)

The dependent variable is VETO, a dummy variable that is 1 if a proposal is vetoed by minority shareholders, and zero if it is passed by minority shareholders. DCAR is a dummy variable equals to one if CAR>0 and zero otherwise. CAR is the market adjusted cumulative abnormal return over the [-2, +10] trading days around the proposal announcement date. MUTUAL_OWN, OTHERINST_OWN and INDIVIDUAL_OWN are defined as in Table 1 except that all of them are measured at the end of the fiscal quarter immediately prior to the proposal voting date. Two-tailed robust p values clustered at the firm level are reported in parentheses in Panel A. The mean marginal effects and z-statistics in Panel C are computed using the formulas in Appendix B.

Panel A. Logit regression results

	Model 1 (main effects model)	Model 2 (interaction effects model)
CONSTANT	-1.5755 (0.000)	-1.7586 (0.077)
DCAR	-0.7994 (0.255)	4.3134 (0.129)
MUTUAL_OWN		0.1160 (0.084)
OTHERINST_OWN		-0.0262 (0.389)
INDIVIDUAL_OWN		-0.1779 (0.544)
DCAR*MUTUAL_OWN		-0.6338 (0.035)
DCAR*OTHERINST_OWN		-0.1720 (0.107)
DCAR*INDIVIDUAL_OWN		-1.0369 (0.313)
Pseudo R-square	0.0227	0.2536

Panel B. Mean marginal effect (mean Z-statistic)

mean marginal effect of DCAR*MUTUAL_OWN	-0.0634 (-1.785)
mean marginal effect of DCAR*OTHERINST_OWN	-0.0145 (-0.444)
mean marginal effect of DCAR*INDIVIDUAL_OWN	-0.0864 (-0.425)

Table 7. The stock market reaction to the announcement of the 2004 regulation

Panel A shows the results of a time-series regression of R_{mt} on EVENT1, EVENT2, $R_{hkm,t-1}$, $R_{hkm,t}$ and $R_{hkm,t+1}$ over a 250-trading day period that ends on December 10, 2004 (i.e., the last trading day of event 2). $R_{m,t}$ is the equally weighted dividend inclusive market return of all A share firms on trading day t. $R_{hkm,t}$ is the equally weighted dividends inclusive market return of all H share and Red Chip share firms that are not listed on the two domestic stock exchanges on trading day t. EVENT1 is a dummy variable equals to 0.2 for a trading day that falls within the [-2, +2] event window centered on event 1, and zero otherwise. EVENT2 is a dummy variable equals to 0.2 for a trading day that falls within the [-2, +2] event window centered on event 2, and zero otherwise. The coefficients in Panels B and C are obtained from a cross-sectional time series regression of individual firm's stock market reaction to EVENT1 and EVENT2 per Sefcik and Thompson (1986). The sample in Panel B includes all A share firms while the sample in Panel C includes only 187 A share firms that are in the same industry (Datastream INDC2) as the H/Red Chip firms and whose total assets at the end of fiscal year 2003 fall between 80% and 120% of the median total assets of the H/Red Chip firms in the same industry. MUTUAL_OW, OTHERINST_OW and INDIVIDUAL_OW are defined as in Table 1 except that they are all measured at the end of the quarter immediately before an event (i.e., event 1 or 2). MV is market value of common shares outstanding, computed as stock price times total shares outstanding at the end of the quarter immediately before an event (i.e., event1 or event2). BM is book-to-market ratio, measured as the book value equity divided by MV.

Panel A. Overall market reaction (N=250 trading days)

Intercept	-0.0002 (0.863)
$R_{hkm,t-1}$	0.0022 (0.964)
$R_{hkm,t}$	0.1774 (0.001)
$R_{hkm,t+1}$	0.0654 (0.288)
Event1	-0.0152 (0.716)
Event2	-0.0269 (0.365)
Adjusted R-square	0.036

Panel B. Cross-sectional variation in the market reaction

	MUTUAL_OW	OTHERINST_OW	INDIVIDUAL_OW	Ln(MV)	Ln(BM)
Event1	0.1978 (0.000)	0.0183 (0.051)	-0.2517 (0.114)	0.0082 (0.087)	0.0067 (0.254)
Event2	0.0887 (0.236)	0.0007 (0.947)	0.0472 (0.556)	0.0107 (0.127)	-0.0049 (0.554)

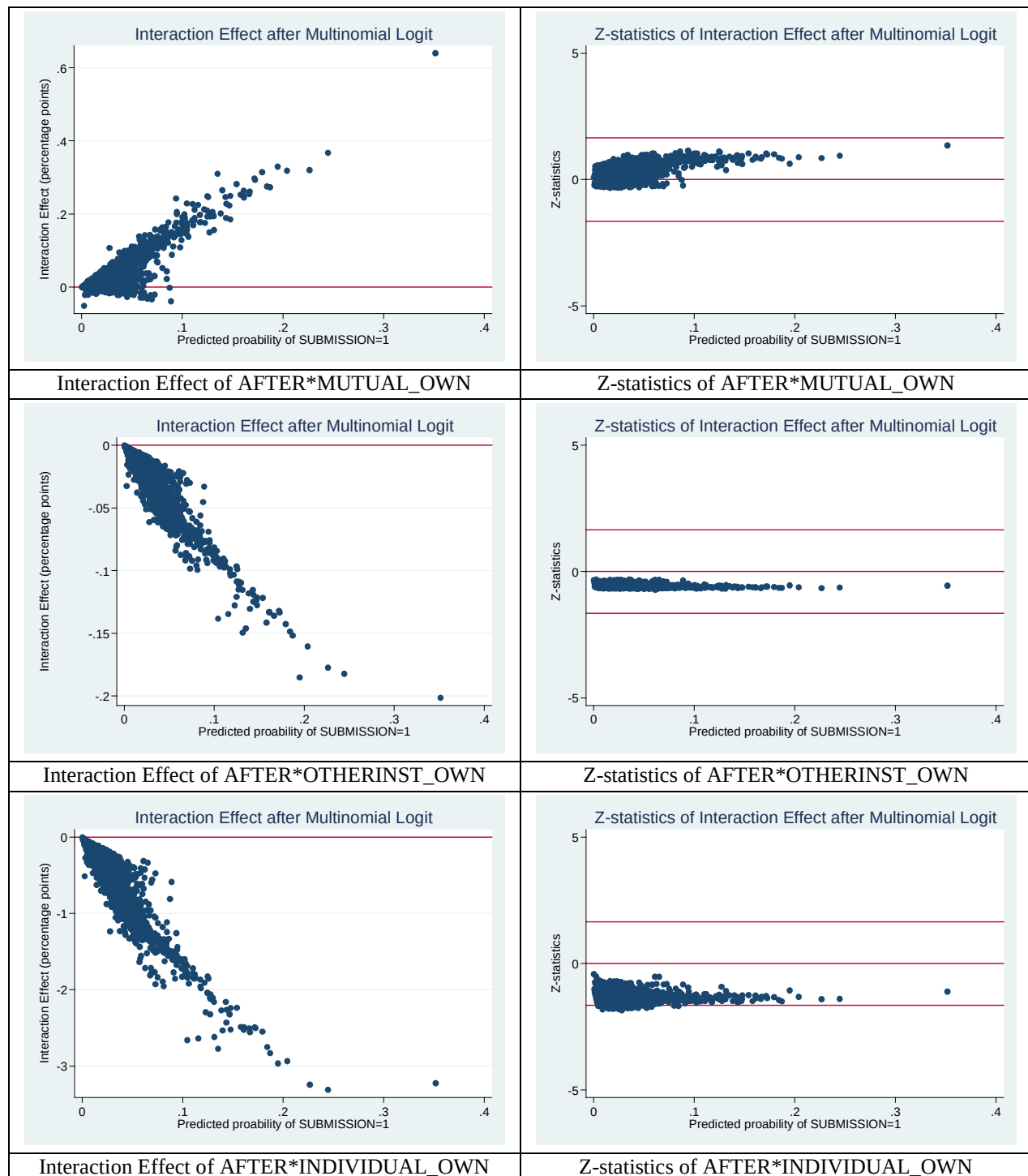
Panel C. A share firms matched with H/Red chip firms by firm size and industry

	MUTUAL_OW	OTHERINST_OW	INDIVIDUAL_OW	Ln(MV)	Ln(BM)
Event1	0.1818 (0.017)	0.0236 (0.163)	-0.0284 (0.768)	0.0086 (0.049)	0.0037 (0.471)
Event2	0.1431 (0.002)	0.0058 (0.751)	0.4897 (0.171)	0.0116 (0.122)	0.0026 (0.699)

Figure 1. The effect of the 2004 regulation on management's incentive to submit value increasing and value decreasing equity offering proposals by the top 10 minority shareholder ownership

The following graphs display the marginal effects and corresponding z-statistics on the interaction variables between AFTER and the top 10 minority shareholder ownership characteristics (MUTUAL_OWN, OTHERINST_OWN, and INDIVIDUAL_OWN) reported in Panel B of Table 2, estimated using the formulas shown in Appendix B. Panel A plots the graphs for the value increasing proposals and Panel B the graphs for the value decreasing proposals. The lines above and below 0 on the figures located on the right side represent the 10% two-tailed significance level (± 1.65).

Panel A. Interaction effects for value increasing proposals (i.e., SUBMISSION=1)



Panel B. Interaction effects for value decreasing proposals (i.e., SUBMISSION=2)

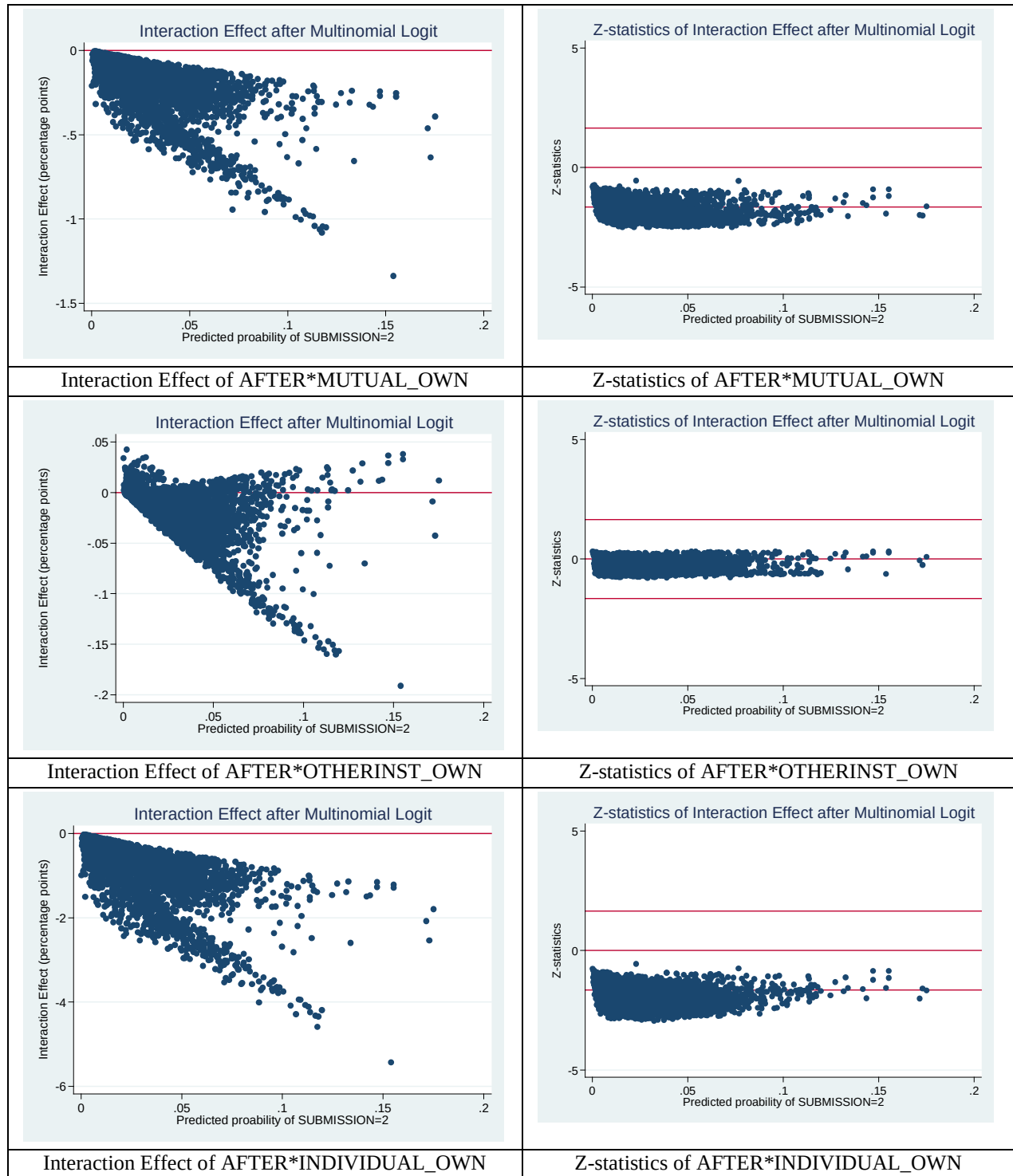


Figure 2. Stock market reactions to equity offering proposal announcements by calendar quarter

See Table 1 for the definition of CAR. The mean/median CAR is computed by calendar quarter. The proposals submitted on or after December 7 in the 4th quarter of 2004 are treated as proposals submitted in the 1st quarter of 2005.

